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Dihydrotanshinone I attenuates estrogen-deficiency bone loss through RANKL-stimulated NF-κB, ERK and NFATc1 signaling pathways

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ABSTRACT

Postmenopausal osteoporosis, a chronic condition that predominantly affects postmenopausal women, presents a significant impediment to their overall well-being. The condition arises from estrogen deficiency, leading to enhanced osteoclast activity. *Salvia miltiorrhiza*, a well-established Chinese herbal medicine with a history of clinical use for osteoporosis treatment, contains diverse active constituents that have shown inhibitory effects on osteoclast formation and bone loss. Dihydrotanshinone I (DTI), a phenanthrenonequinone compound derived from the root of *Salvia miltiorrhiza*, has been identified as a potential therapeutic agent, although its mechanism of action on osteoclasts remains elusive.

In this study, we aimed to elucidate the inhibitory potential of DTI on RANKL-induced osteoclastogenesis. We observed the ability of DTI to effectively impede the expression of key osteoclast-specific genes and proteins, as assessed by Real-time PCR and Western Blotting analyses. Mechanistically, DTI exerted its inhibitory effects on osteoclast formation by modulating critical signaling pathways including NF-κB, ERK, and calcium ion signaling. Notably, DTI intervention disrupted the nuclear translocation and subsequent transcriptional activity of the NFATc1, thus providing mechanistic insights into its inhibitory role in osteoclastogenesis. To further assess the therapeutic potential of DTI, we employed an ovariectomized osteoporosis animal model to examine its impact on bone loss. Encouragingly, DTI demonstrated efficacy in mitigating bone loss induced by estrogen deficiency.

In conclusion, our investigation elucidates the ability of DTI to regulate multiple signaling pathways activated by RANKL, leading to the inhibition of osteoclast formation and prevention of estrogen-deficiency osteoporosis. Consequently, DTI emerges as a promising candidate for the treatment of osteoporosis.

1. Introduction

Bone is a dynamic organ in a long-term process of self-renewal and remodeling. Under the synergistic function of osteoblasts and osteoclasts, bone microstructure remodeling is completed. Postmenopausal osteoporosis(PMO) is a common bone metabolic disease, the lack of estrogen causes the balance mechanism of bone resorption-formation to be broken, and the function of osteoclasts is hyperactive, resulting in bone loss and being prone to fractures, the annual cost of osteoporosis-related fragility fractures poses a substantial economic burden [1,2]. However, despite the promising results of clinically available postmenopausal osteoporosis treatments including bisphosphonates, and estrogen supplementation, some side effects have also been reported, such as cancer [3], osteonecrosis [4]] and heart disease [7,6], highlighting the need for safer alternative treatments.

Osteoclast differentiation is primarily regulated by the sequential actions of Macrophage Colony-Stimulating Factor (M-CSF) and Receptor Activator of Nuclear Factor-κB Ligand (RANKL) [7,8]. M-CSF stimulation induces the expression of RANK on the surface of osteoclast precursors. Upon binding of RANK with RANKL, a series of intracellular signaling events are initiated, including the activation of NF-κB, mitogen-activated protein kinases (MAPKs), and calcium oscillations [9]. These events collectively promote the nuclear import of the transcription factor Nuclear factor of activated T cell cytoplasmic 1 (NFATc1), which is critically involved in osteoclastogenesis. Subsequently, the translocation and amplification of NFATc1 within the

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nucleus facilitate the expression of osteoclast-specific genes and proteins [10,11]. Notably, the calcium-mediated calcineurin pathway tightly regulates the nuclear translocation and activation of NFATc1 [11].

In recent years, there has been a notable shift towards the investigation of safer and more efficacious compounds or extracts derived from traditional Chinese medicine. The identification and screening of natural products with potential anti-osteoporosis properties have emerged as a prominent area of research [12]. Numerous traditional Chinese medicine ingredients have exhibited remarkable anti-osteoporosis effects [13–16]. Among these, Dihydrotanshinone I holds particular significance as a naturally occurring bioactive compound extracted from the rhizome of *Salvia miltiorrhiza*, a medicinal plant widely employed in orthopedic and cardiovascular disciplines. *Salvia miltiorrhiza* has been extensively utilized in clinical trials evaluating the efficacy of herbal remedies for osteoporosis, with more than 30% of prescribed treatments incorporating this herb. *Salvia miltiorrhiza* encompasses diverse bone-targeting natural active ingredients, some of which have demonstrated anti-osteoclastogenesis activity; however, the precise underlying mechanism remains unknown [17]. Consequently, the active constituent DTI possesses a stronger clinical foundation for therapeutic applications compared to other traditional Chinese medicine ingredients. DTI has been reported to exhibit antibacterial effects [18], induce apoptosis in tumor cells [19, 20], inhibit angiogenesis [21,22] and mitigate myocardial ischemia–reperfusion injury [23,24]. Additionally, DTI has been shown to impede aortic valve interstitial cell calcification through the NF-κB/ERK pathway [25].

In this investigation, we explored the inhibitory properties of DTI on the differentiation of RAW264.7 cells into osteoclasts in *vitro*. Our study demonstrated the ability of DTI to effectively suppress the activation of crucial signaling pathways involved in osteoclastogenesis, namely NF-κB, ERK and NFATc1. Specifically, DTI exhibited inhibitory effects on NFATc1 activity and its translocation into the nucleus, thereby impeding the maturation of osteoclasts. Moreover, we assessed the therapeutic potential of DTI for mitigating bone loss using an ovariectomized mouse model. The results indicated that DTI possessed the capability to attenuate bone loss in this experimental model. These findings lay a preliminary groundwork for future clinical investigations, suggesting that dihydrotanshinone I holds promise as a potential therapeutic agent for managing osteoporosis.

2. Materials and methods

2.1. Materials and regents

Dihydrotanshinone I (purity > 98%), obtained from Wuhan ChemFaces company, was solubilized in DMSO to achieve a storage concentration of 5 mM. The solution was stored at −80 °C and subsequently diluted with α-MEM to various concentrations as needed. Thermo Fisher Scientific provided the following reagents: fetal bovine serum (FBS) and α-MEM. RANKL, purchased from R&D Corporation, was dissolved in PBS and stored at −80 °C. The Cell Counting Kit-8 was commercially obtained from Dojindo. Primary antibodies specific to c-Fos, p-ERK, JNK, p-JNK, NF-kappa-B inhibitor alpha (IκBα), p65, NFATc1, CTSK, ERK, and β-actin were sourced from Santa Cruz or Cell Signaling Technology. Proteintech (Wuhan, China) provided the anti-mouse or anti-rabbit secondary antibodies used in this study.

2.2. Cell culture and cytotoxicity test

The RAW264.7 cell line, obtained from Procell Life Science&Technology Company (Wuhan, China), was cultured in α-MEM media supplemented with 10% FBS in Petri dishes. Subsequently, the RAW264.7 cells were seeded into the designated culture plates. The following day, the culture medium was replaced with varying concentrations of DTI, and the cells were co-cultured for a duration of 48 h. To assess cell viability, the absorbance at 450 nm was measured, following by adding 10 μl of CCK8 kit to each well and incubating for an additional 2 h.

2.3. Osteoclastogenesis assay and TRAcP staining

RAW264.7 cells in a healthy state were seeded onto plates at a density of $2.5x10^3$ cells per well after overnight adherence. On the following day, the culture medium was switched to one containing RANKL at a concentration of 10 ng/mL. Different concentrations of DTI (0, 0.5, 1, 1.5, and 2uM) were added to the medium to intervene in the osteoclastogenesis process. The medium was refreshed every 48 h until the fifth day when osteoclast maturation was observed. The cells were subsequently fixed with 4% PFA, washed with PBS, and treated with TRAcP staining solution. Incubation in a 60-degree incubator for 30–60 min was performed. Cells exhibiting three or more nuclei under a light microscope were characterized as osteoclasts.

2.4. Podosome belt staining

The RAW264.7 cells were seeded onto plates as described previously. After overnight attachment, RANKL was introduced to induce osteoclast differentiation. Simultaneously, different concentrations of DTI (1uM, 2uM) were administered for treatment. Upon observation of osteoclast maturation on the fifth day, the cells were fixed using 4% PFA for 1 h. Following that, the cell membrane underwent permeabilization with 0.5% Triton X-100 solution for 30 min, followed by a blocking step utilizing 10% bovine serum albumin (BSA) for 2 h at room temperature. Rhodamine phalloidin was then applied to stain the podosome belt under dark conditions, while the nuclei of osteoclasts were stained with DAPI. Finally, fluorescence microscopy was employed for subsequent observation and capturing of images.

2.5. RNA extraction and Real-time PCR

The RAW264.7 cells were inoculated on plates at a density of 8×10^4 cells per well under the stimulation of RANKL, until the osteoclasts matured. 1 ml Trizol per well was added to lyse the cells to obtain total RNA which was reversed into cDNA, following the manufacturer's instructions for subsequent steps. Amplification was performed on Applied Biosystems. The relevant primer sequences were shown below:

Genes	Forward(5′-3′)	Reverse(5′-3′)
Nfatc1	CAACGCCCTGACCACCGATAG	GGCTGCCTTCCGTCTCATAGT
Atp6v0d2	GTGAGACCTTGGAAGACCTGAA	GAGAAATGTGCTCAGGGGCT
Ctsk	GGGAGAAAAACCTGAAGC	ATTCTGGGGACTCAGAGC
c-Fos	GCGAGCAACTGAGAAGAC	TTGAAACCCGAGAACATC
Acp5	TGTGGCCATCTTTATGCT	GTCATTTCTTTGGGGCTT
β-actin	GGCTGTATTCCCCTCCATCG	CCAGTTGGTAACAATGCCATGT

2.6. Western blot assay

The RAW264.7 cells were seeded onto plates containing 10 ng/ml RANKL in the presence or absence of DTI in the medium. Following cell intervention, the cells were lysed using RIPA lysate (Beyotime, China). Cellular proteins were then collected and separated by electrophoresis on a 4%-12% gel. Afterward, the cellular proteins were transferred onto a PVDF Transfer Membrane. The membrane was subjected to blocking using skimmed milk at room temperature for a duration of 2 h, followed by incubation with specific primary antibodies (diluted at 1:1000) overnight at 4 °C on a rotary shaker. The following day, the membrane was washed and incubated with the secondary antibody (diluted at 1:4000) at room temperature for 1 h. Finally, the membrane was incubated with an ultra-sensitive ECL chemiluminescent reagent (Beyotime, China) for 1 min, and the images were captured using the Bio-Rad gel imaging system. Image J software was utilized for subsequent analysis.

2.7. NFATc1 fluorescence detection

The RAW264.7 cells were seeded onto plates, and the following day, RANKL was added to induce the differentiation, with or without the presence of DTI (2uM). The medium was changed every 48 h. After the fifth day of osteoclast maturation, the cells were fixed using 4% PFA and subjected to three washes with PBS. Subsequently, the cell membrane was permeabilized using 0.5% Triton X-100 and blocked with 10% BSA at room temperature for 2 h. The cells were then exposed to the NFATc1 primary antibody, which was incubated overnight at 4 °C. Following this, the cells were incubated with a 488-labeled secondary antibody (Invitrogen, USA) for 1 h in a dark environment. Additionally, rhodamine phalloidin was applied for an extra 30 min, followed by DAPI staining for 20 min. The cells were immediately observed and imaged using a fluorescence microscope.

2.8. Calcium oscillation assay

The RAW264.7 cells were seeded onto plates as previously described. On the following day, the cells were subjected to a pre-treatment with DTI (2uM) for 2 h, followed by incubation with RANKL for 24 h. Subsequently, the cells were washed three times with HANKS balanced salt solution supplemented with 1% FBS and 250 mM probenecid. The cells were then incubated with Fluo-4 AM solution (Biosharp, China), containing 0.02% Pluronic F-127, at 37 °C in the dark for 30 min. After three additional washes with HANKS balanced salt solution, the fluorescence intensity was immediately measured using flow cytometry.

2.9. p65 nuclear translocation

The RAW264.7 cells were plated and allowed to adhere overnight. Following this, a pre-treatment step was performed, exposing the cells to DTI for 2 h. Subsequently, RANKL was introduced into the culture and incubated for an additional hour. The cells were fixed using 4% PFA. To permeabilize the cell membrane, 0.5% Triton X-100 was applied for 1 h. Subsequently, the cells underwent blocking with 10% BSA at room temperature for2 h. Following the blocking step, the cells were subjected to overnight incubation with a primary antibody specifically targeting p65. After thorough washing of the cells with PBS three times, a secondary antibody labeled with Cy5 was applied and incubated at room temperature in a dark environment for 1 h. DAPI staining was performed for 20 min to visualize the nuclei. Finally, the cells were immediately observed using a laser confocal microscope.

2.10. Ovariectomised (OVX) mouse model

Following approval by the Experimental Animal Ethics Committee of Guangzhou University of Chinese Medicine (No. 20220622003), C57BL/6J female mice at 8 weeks of age were randomly divided into three groups. The groups were designated as follows: sham operation group, OVX (ovariectomy) with vehicle administration group, and OVX with DTI intervention group. The mice in the OVX group underwent bilateral oophorectomy under anesthesia induced by intraperitoneal injection of pentobarbital sodium, while the sham operation group underwent bilateral ovary exposure without oophorectomy. Following a recovery period of one week post-surgery, the DTI treatment group received intraperitoneal injections of DTI at a dose of 10 mg/kg every 2 days for a duration of 6 weeks. The remaining mice were administered an intraperitoneal injection of an equivalent volume of PBS. At the conclusion of all procedures, all mice were sacrificed, and femurs were collected for further analysis.

2.11. MicroCT scanning and analysis

The right femur underwent meticulous removal of superficial soft tissue and was immersed in 4% paraformaldehyde solution for 48 h, then transferred to PBS for storage purposes. Subsequently, the femur was subjected to scanning using a Skyscan 1172 MicroCT scanner (Skyscan, Bruker, Belgium). The region of interest (ROI) for analysis was defined as the bone trabeculae within the range of 0.5–1.5 mm above the femoral growth plate. The scanning parameters remained consistent with previous studies [26]. The bone tissue data, including bone mineral density (BMD), bone volume/tissue volume (BV/TV), trabecular number (Tb.N), trabecular thickness (Tb.Th), and trabecular separation (Tb.Sp), were analyzed using the Bruker CTAn software (Skyscan, Belgium).

2.12. Bone histomorphometric analysis

Following MicroCT scanning, the femur samples were subjected to immersion in 14% ethylenediaminetetraacetic acid (EDTA) solution on a shaking table at room temperature. The femur samples were treated with fresh EDTA solution every 2 days until achieving a softened state. Following dehydration with ethanol, the samples were meticulously embedded in paraffin and cut into slices of 5 μm thickness. Subsequently, the sections were subjected to staining with hematoxylin and eosin (HE) and TRAcP to facilitate examination and analysis. The stained sections were scanned using CaseViewer 2.4 (3DHISTECH Ltd.), and bone histomorphology was analyzed using Image J software.

2.13. Statistical analysis

All statistical graphs were expressed using mean ± SEM, and differences between groups were analyzed using one-way ANOVA. p values < 0.05 were considered significant differences. All statistics were performed using GraphPad Prism 9.0 software.

3. Results

3.1. DTI inhibited RANKL-induced osteoclastogenesis in vitro

To investigate the impact of DTI (Fig. 1A) on osteoclastogenesis, RAW264.7 cells were initially cultured for 48 h in the presence or absence of DTI, followed by a CCK8 assay. Absorbance measurements indicated that DTI exhibited no significant cytotoxicity towards RAW264.7 cell proliferation within the concentration range of 0.5-2uM (Fig. 1B). Subsequently, DTI was added to the culture medium in a dose-dependent manner during RANKL-induced osteoclastogenesis. Remarkably, escalating concentrations of DTI exerted inhibitory effects on TRAcP-positive osteoclast formation, with the lowest osteoclast count observed at 2uM (Fig. 1C-D). Notably, the pseudopodia band, a distinctive feature associated with osteoclast function, was subjected to fluorescent staining. This analysis revealed a dose-dependent decline in osteoclast formation at concentrations of 1uM and 2uM, as evidenced by reductions in the average area and the number of nuclei per osteoclast (Fig. 1E-F).

3.2. DTI reduced RANKL-induced expression of osteoclast-specific markers

After subjecting the osteoclastogenesis process to RANKL stimulation and DTI intervention at different time points, we conducted Real-time PCR and Western Blot analyses to assess the amplification and expression of osteoclast-specific genes and proteins. The PCR results clearly demonstrated that DTI treatment led to a noticeable decrease in the expression of osteoclast-specific genes, namely *Acp5*, *c-Fos*, *Ctsk*, and *Atp6v0d2* (Fig. 2A). Correspondingly, the Western Blot findings revealed a consistent pattern, where the administration of DTI resulted in reduced levels of secreted c-Fos protein on days 1 and 5, as well as a significant down-regulation in the expression of CTSK, a protein associated with bone resorption, on day 5 (Fig. 2B-C). These results aligned with the observed gene amplification outcomes, providing further support for the inhibitory effects of DTI on osteoclast formation.

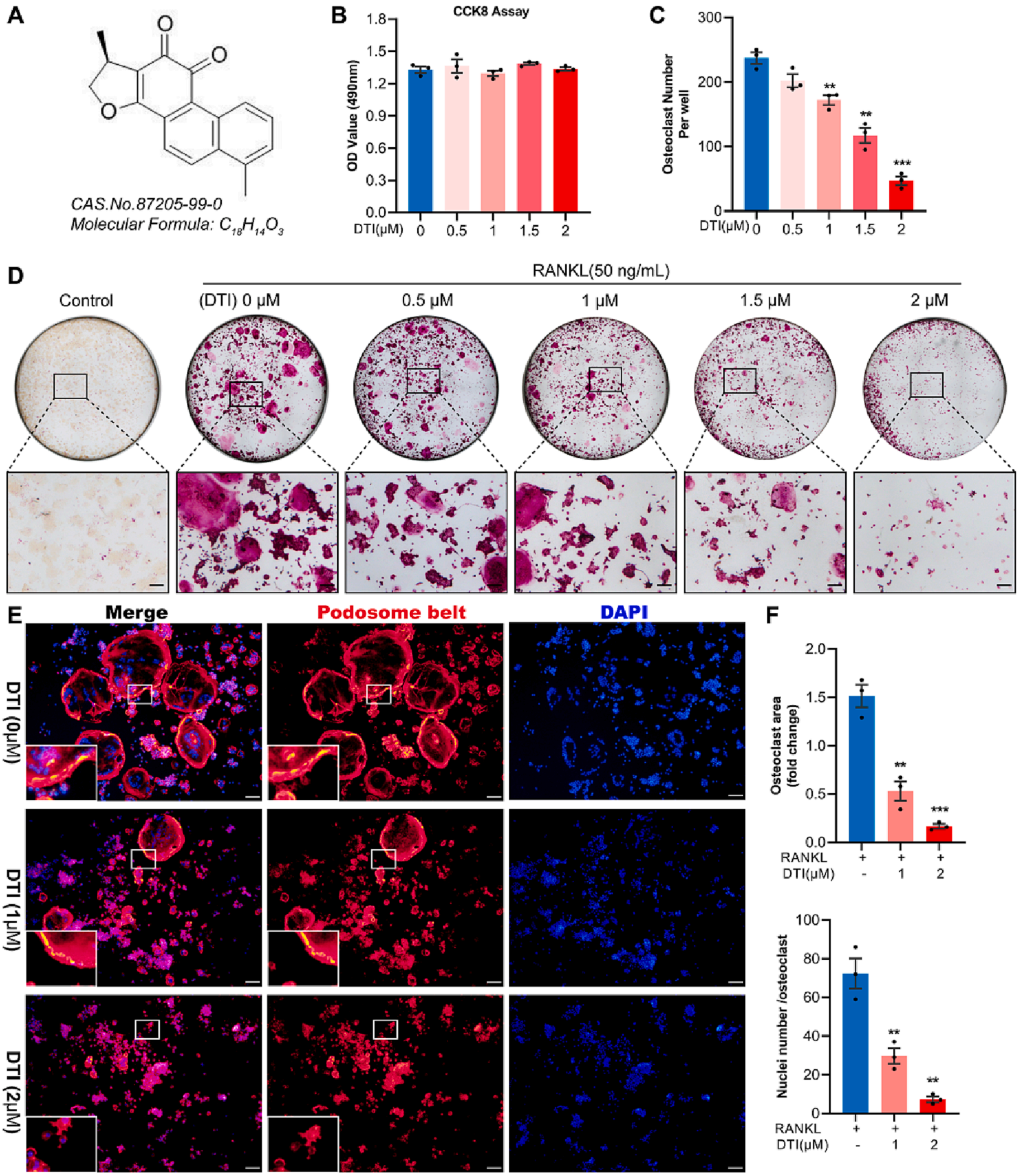

Fig. 1. DTI inhibited RANKL-induced osteoclastogenesis in vitro. A. Molecular structure and chemical formula of DTI. B. CCK8 test results showed no significant toxic effect on the proliferation of RAW264.7 cells after DTI intervention for 48 h. C. Osteoclast counts showed that fewer osteoclasts were observed with higher DTI concentrations. D. Representative pictures of TRAcP stained osteoclasts after different concentrations of intervention. E. Phalloidin (red) labeled pseudopodia bands and DAPI (blue) labeled nuclei after intervention with different concentrations of DTI. Scale bar = 200 μm. F. Quantitative analysis of the area of the pseudopodia band and the number of osteoclast nuclei. Data was displayed with histograms: n = 3. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

3.3. DTI inhibited RANKL-induced MAPK and NF-κB signaling pathways

The process of osteoclastogenesis is closely associated with the NF-κB and MAPK signaling pathways. Activation of the NF-κB pathway is characterized by the degradation of IκBα protein, while ERK, JNK and p38 are key members of the MAPK family whose phosphorylation levels are increased in the presence of RANKL, thereby regulating osteoclast differentiation. To elucidate the potential molecular mechanism of DTI in osteoclastogenesis, we conducted a short-time gradient RANKL stimulation and DTI intervention experiment to investigate the involvement of the NF-κB and MAPK signaling pathways. Western blot results demonstrated that DTI inhibited the phosphorylation of ERK at 10 min and 20 min, while the phosphorylation of JNK was not significantly affected (Fig. 3A-B). Furthermore, DTI effectively reversed the degradation of IκBα protein at 30 min and 60 min. Immunofluorescence experiments further revealed that DTI treatment prevented the

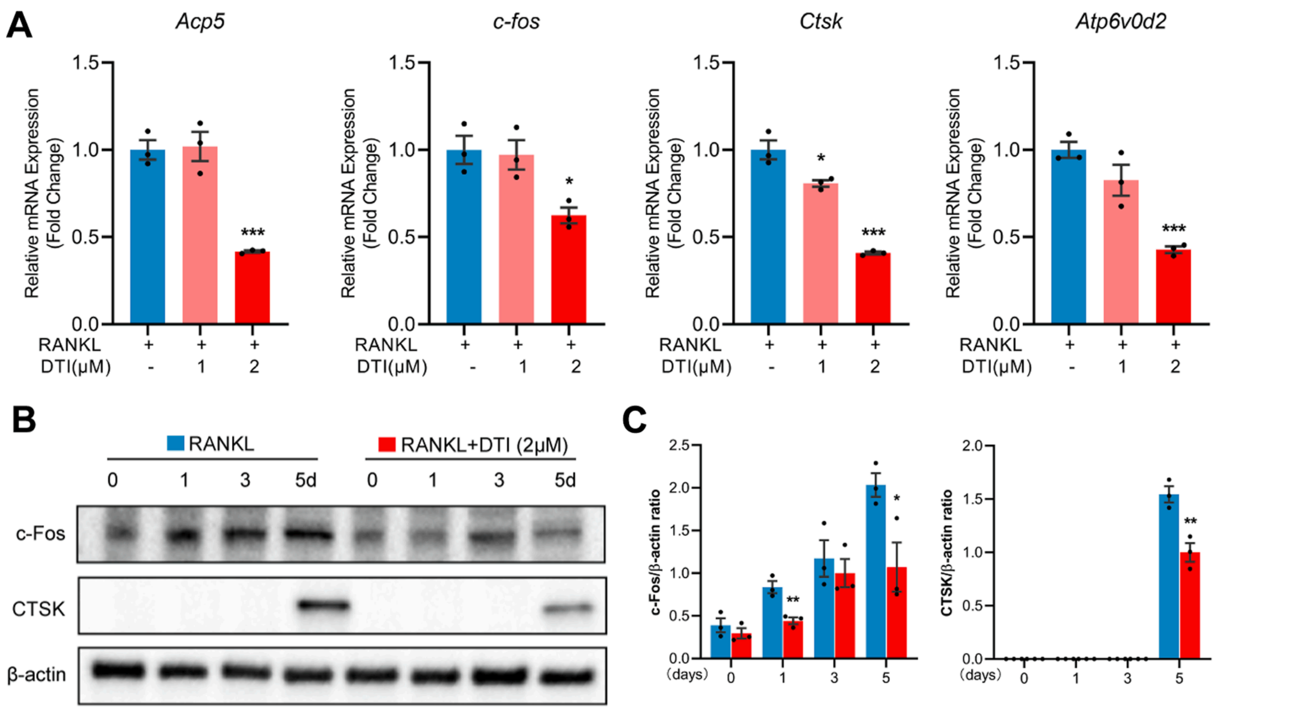

Fig. 2. DTI reduced the secretion of osteoclast-specific markers. A. DTI at a concentration of 2uM significantly inhibited the amplification of osteoclast-specific genes. B. DTI significantly decreased the expression of c-Fos and CTSK. n = 3. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

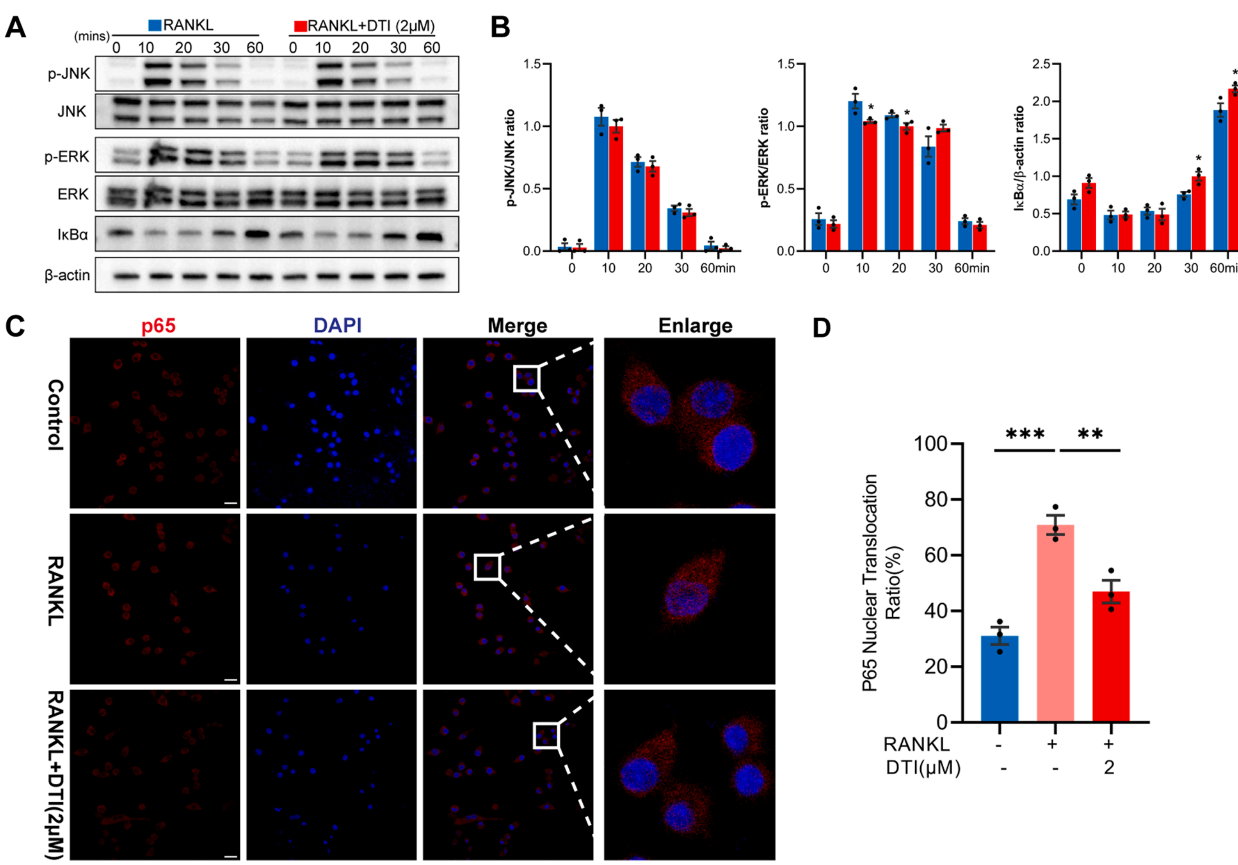

Fig. 3. Effect of DTI on MAPK and NF-κB signaling pathway marker proteins. A. DTI inhibited the phosphorylation of ERK at 10 min and 20 min but had no obvious effect on JNK. DTI reversed the degradation of IκBα protein at 30 min and 60 min. . B. Quantitative Analysis of Western Blot Results. C. Representative images of nuclear heterotopia of p65 upon RANKL stimulation with and without DTI. Scale bar = 200 μm. D. Quantification of p65 nuclear translocation ratio. n = 3. *$p < 0.05$, **$p < 0.01$.

translocation of p65 into the nucleus, corroborating the findings from the immunoblotting analysis (Fig. 3C-D).

3.4. DTI mitigated RANKL-induced calcium ion oscillation and NFATc1 activity

Calcium ion levels are closely related to the outcome of the expansion and translocation of NFATc1 to the nucleus, and calcium ions-induced oscillations under RANKL stimulation promote the expression and transfer of NFATc1. RAW264.7 cells loaded with Fluo-4 AM fluorescent probe were used to detect the effect of DTI on calcium ion oscillation induced by RANKL. Flow cytometry results revealed that DTI can inhibit the Calcium ion fluorescence intensity stimulated by RANKL (Fig. 4 E-F). Consistent with downregulated calcium, DTI inhibited NFATc1 protein and gene expression, with the most pronounced reversal at day 5(Fig. 4 A-D).

3.5. DTI prevented bone mass loss in OVX mice

The mice were injected intraperitoneally with the vehicle and DTI at a dose of 10 mg/kg for 6 weeks, once every 2 days and their body weight was recorded once a week, lowercase death or obvious adverse reactions were observed during the whole experiment (Fig. 5 A-B). MicroCT results showed that DTI could prevent bone loss caused by estrogen

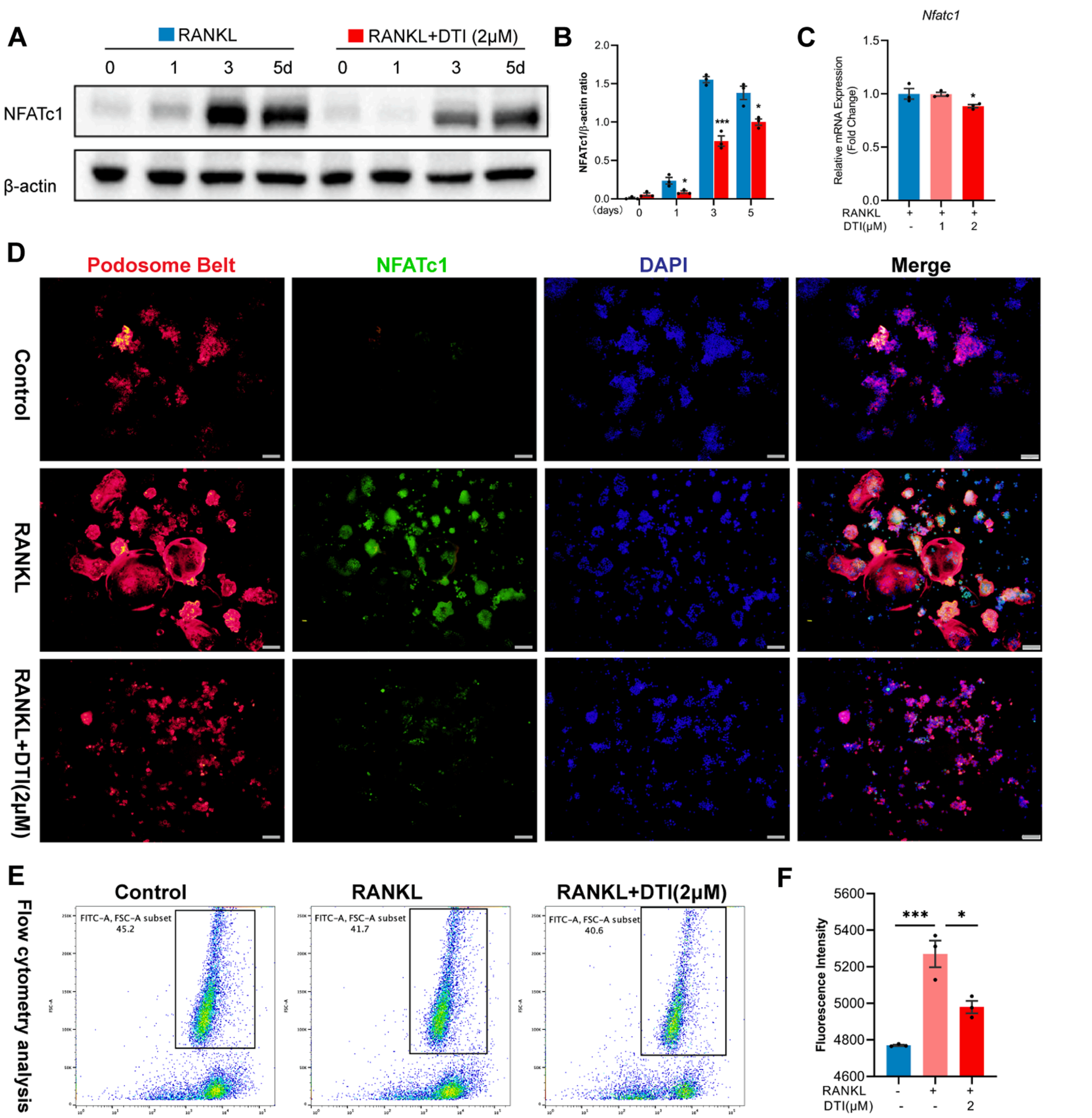

Fig. 4. DTI can reduce calcium ion oscillation level and NFATc1 activity. A. DTI (2uM) can significantly inhibit the expression of NFATc1 protein. B. Quantitative analysis of the NFATc1 western blot. C. DTI inhibits the amplification of NFATc1 gene. D. DTI blocks the secretion and nuclear import of NFATc1 protein in a concentration-dependent manner. Scale bar = 200 μm. E. DTI reduces the calcium ion oscillation level induced by RANKL. F. Quantitative analysis of FITC-labeled calcium ion fluorescence intensity. n = 3. *p < 0.05, **p < 0.01, ***p < 0.001.

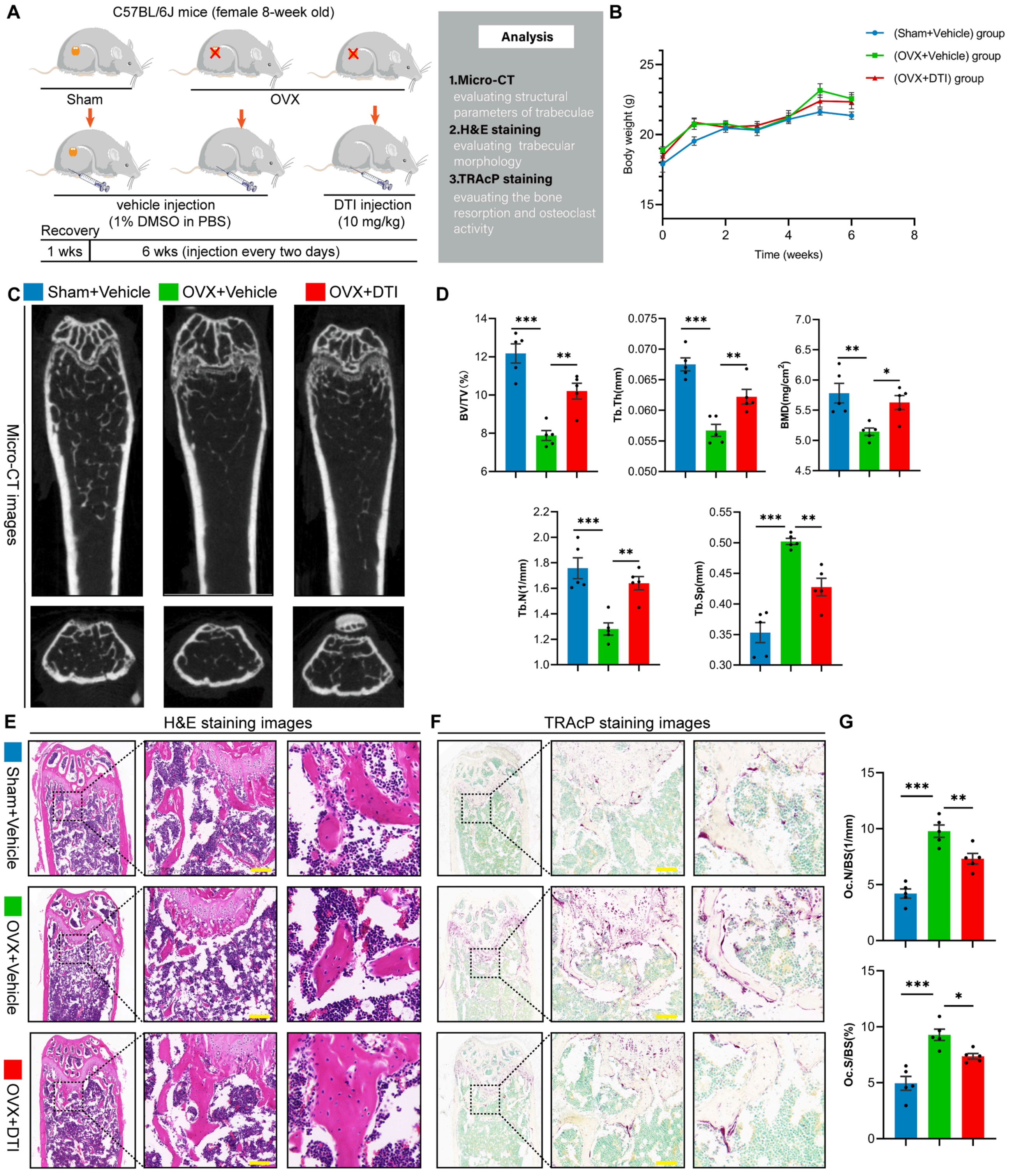

Fig. 5. DTI can reduce bone loss in OVX mice A. DTI was injected intraperitoneally at a 10 mg/kg dose for 6 weeks. B. Body weight changes during DTI intervention of mice in each group. C. Mouse distal femur images after Micro CT reconstruction. D. DTI can reverse bone loss in OVX mice, manifested in changes in bone microstructure, including quantification of bone parameters such as BV/TV, Tb.Th, Tb.N, Tb.Sp and BMD. E. Representative images of HE staining of femur sections. Scale bar=200 μm. F. Representative images of TRAcP staining of femur sections. G. Quantification of osteoclast number and surface area in bone slices. n = 5. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

deficiency, compared with OVX group mice. After DTI treatment, bone microstructural parameters including BV/TV, Tb.Th, Tb.N, Tb.Sp and BMD were reversed (Fig. 5 C-D). Further histomorphological results also confirmed that DTI can maintain the integrity of bone structure, HE staining showed that the surface and integrity of trabecular bone were well preserved after DTI treatment. TRAcP staining showed that estrogen deficiency caused the hyperactivity of osteoclasts in OVX mice, while DTI could inhibit the activity of osteoclasts on the surface of trabecular bone (Fig. 5 E-G).

4. Discussion

Postmenopausal osteoporosis, a commonly encountered chronic disorder, is characterized by an augmented susceptibility to fractures attributable to the excessive activity of osteoclasts and an impaired equilibrium in the dynamic process of bone remodeling. These detrimental effects stem from the deficiency of estrogen following menopause [1,27]. Although current treatment approaches for postmenopausal osteoporosis exist, they often entail potential side effects. For instance, the administration of exogenous estrogen to supplement the hormonal deficiency in postmenopausal women has been linked to an increased incidence of breast cancer [3] and cardiovascular events . Consequently, there is a growing interest in safer and natural alternative therapies, which have emerged as a focal point in the prevention and treatment of osteoporosis. In this study, we conducted in vitro experiments to investigate the effects of DTI on osteoclast formation. Our study provides evidence that DTI exerts inhibitory effects on osteoclastogenesis by modulating RANKL-induced signaling pathways, such as NF-κB, ERK, and NFATc1, leading to a reduction in the expression of osteoclast-specific genes (Fig. 6). Additionally, we confirmed the therapeutic efficacy of DTI in ameliorating bone loss in a mouse model of estrogen deficiency, highlighting its potential as a therapeutic intervention for osteoporosis.

Osteoclasts, which originate from macrophages, undergo a maturation process facilitated by M-CSF and RANKL. In the presence of these factors, osteoclasts gradually develop into multinucleated cells and assemble pseudopodia body bands on non-mineralized matrices. We first demonstrated that the effect of DTI on inhibiting osteoclast formation was related to the concentration. The Pseudopodia band is a landmark structure of osteoclast bone resorption [28]. From immunofluorescence staining, we found that DTI significantly inhibited the formation of pseudopod bands.

Upon contacting between RANKL and its corresponding receptors on osteoclast precursor cells, multiple downstream signaling pathways are triggered. Notably, the NF-κB and MAPK signaling pathways are pivotal in the process of osteoclastogenesis [8]. Following RANKL stimulation, IκBα undergoes degradation, and p65 translocates to the nucleus. The Western blot analysis, employing short-time gradients, revealed that DTI administration resulted in the reversal of IκBα degradation at 30 min and 60 min, indicating its inhibitory effect on NF-κB activity. Immunofluorescence assessment of p65 localization corroborated these findings, demonstrating that DTI impedes the translocation of p65 into the nucleus. Additionally, extracellular signal-regulated kinase (ERK) and c-Jun N-terminal kinase (JNK), prominent members of the MAPK family, are crucial in regulating osteoclastogenesis and bone resorption in response to RANKL stimulation . Our immunoblotting results demonstrated that DTI effectively suppressed the expression of p-ERK1/2 induced by RANKL at 10 min and 20 min, while exhibiting no substantial inhibitory effect on JNK phosphorylation.

NFATc1 is the primary regulator of osteoclasts. During the differentiation and maturation of osteoclasts, NFATc1 begins to express under the stimulation of RANKL, reaches a peak around day 3, and can promote the expression of multiple bone resorption-related genes and proteins. Our data showed that DTI effectively reduced NFATc1 protein and gene amplification and expression. The fluorescent staining results were consistent with Western blot, further confirming that DTI can effectively prevent its expression and transfer into the nucleus. This effect led to the downregulation of NFATc1-regulated osteoclast unique markers, including *Ctsk*, *c-Fos*, *Acp5* and *Atp6vod2*. In addition, calcium ion oscillations increased when RANKL exists, which promoted the activation of calcineurin protein, thereby promoting the expression and self-amplification of NFATc1 [11]. It is worth noting that DTI also can inhibit the fluorescence intensity of calcium ion oscillation induced by RANKL, which was detected through flow cytometry. These results suggest that DTI can inhibit NFATc1-mediated osteoclast activity and the secretion of bone resorption-related markers.

Based on the previously discussed effects of DTI on osteoclast formation and function, as well as its impact on RANKL-mediated signaling pathways, we developed an osteoporosis mouse model following ovariectomy. Consistently, our MicroCT analysis revealed the efficacy of DTI in effectively mitigating bone loss induced by estrogen deficiency, as evidenced by significant improvements in BV/TV, Tb.Th, BMD, Tb.N,

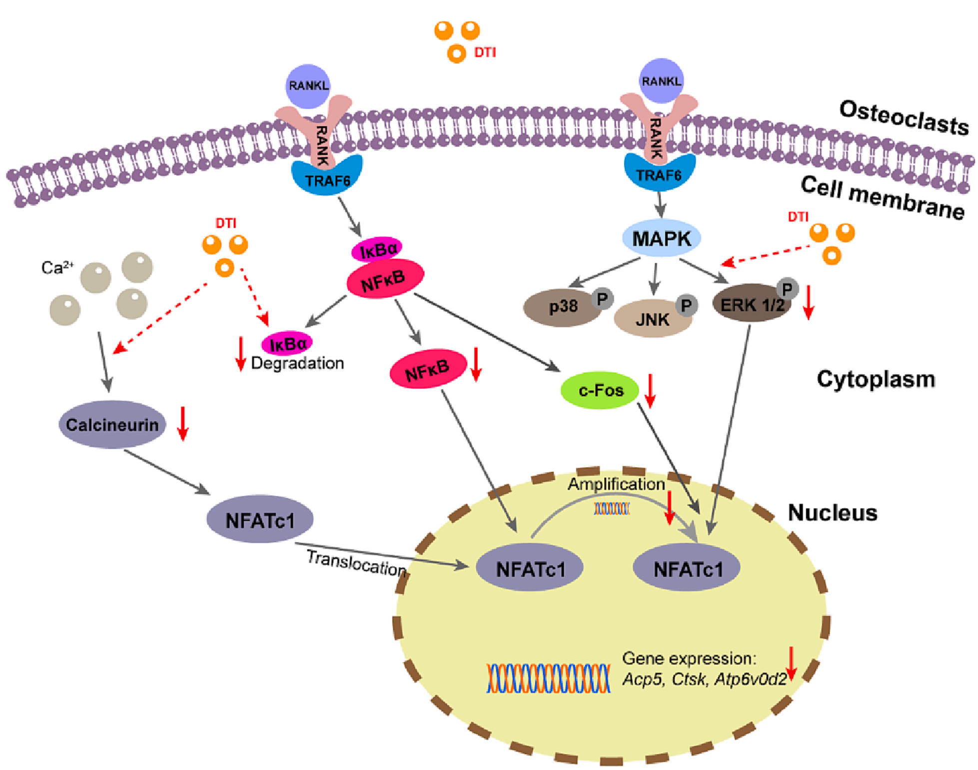

Fig. 6. Schematic diagram of the impact of DTI on RANKL-stimulated signaling pathways.

and Tb.Sp parameters. Furthermore, histological examinations through HE and TRAcP staining demonstrated the ability of DTI to effectively suppress osteoclast activity in the vicinity of trabecular bone, thereby preserving the integrity of trabecular architecture. Taken together, these findings support the potential of DTI as a promising natural therapeutic agent for the treatment of osteoporosis.

CRediT authorship contribution statement

Chao Ma: Conceptualization, Validation, Investigation, Writing – original draft. **Liang Mo:** Investigation, Formal analysis. **Zhangzheng Wang:** Investigation, Formal analysis. **Deqiang Peng:** Data curation. **Chi Zhou:** Resources. **Wei Niu:** Project administration, Funding acquisition. **Yuhao Liu:** Project administration, Funding acquisition, Writing - review & editing. **Zhenqiu Chen:** Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intimp.2023.110572.

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